September 25, 2026

Agenda

- Motivation for long-range, low-power networks
- Non-3GPP solutions: LoRaWAN, Sigfox, Weightless, RPMA, EnOcean and Z-Wave
- 3GPP solutions: LTE-M and NB-IoT
- The role of 5G in supporting the Internet of Things
- Cost analysis of communications
- Evolution prospects

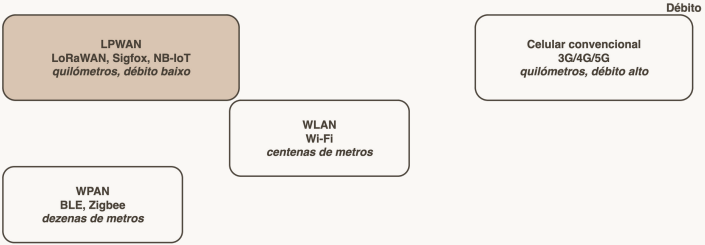
Why Long-Range, Low-Power Networks (1/2)

- The short-range technologies covered in the previous lesson do not reach scenarios spread over several kilometres, such as agriculture, urban utility metering or environmental monitoring.
- Conventional cellular networks do cover such distances, but with energy consumption and per-device cost incompatible with deployments of thousands of battery-powered sensors.

Why Long-Range, Low-Power Networks (2/2)

- LPWAN (*Low-Power Wide Area Network*) technologies address this gap: they favour range and battery life, accepting very low data rates and high latency in return.
- Two distinct families occupy this space: non-3GPP solutions, operating in unlicensed spectrum, and 3GPP solutions, integrated into licensed mobile networks.

Positioning LPWAN in the Solution Space



LPWAN ocupa o espaço de alcance elevado com débito e consumo baixos

LoRa and LoRaWAN: Background

- LoRa was developed by Cycleo in 2010 and later acquired by Semtech, which developed the LoRaWAN network.
- The LoRa Alliance was created in 2015 to promote and standardise the technology.
- Amsterdam was the first city covered by a LoRa network.
- In June 2015, Bouygues Telecom announced coverage of the entire French territory by a LoRaWAN network, completed by the end of 2016.
- The term LoRa designates the physical layer, that is, the modulation technique; LoRaWAN designates the network protocol built on top of it.

LoRaWAN: Technical Characteristics (1/2)

- Chirp Spread Spectrum modulation, with a channel bandwidth typically of 125 kHz.
- High reception sensitivity: down to -137 dBm at end devices and -142 dBm at gateways.
- Long range, reaching up to 15 km under favourable conditions.
- Good indoor penetration: with high spreading factors, the additional margin can reach 20 dB.

LoRaWAN: Technical Characteristics (2/2)

- Robustness against channel noise, since the signal occupies the whole available bandwidth during transmission.
- Resistance to the Doppler effect, to multipath propagation and to signal fading.

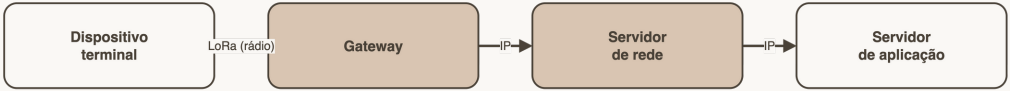
LoRaWAN: Architecture and Device Classes (1/2)

- The architecture is organised as a star: end devices communicate with LoRa gateways, which forward traffic to a network server and, from there, to the customer application server.
- Any LoRa device can both transmit and receive, unlike strictly unidirectional technologies.
- Payload per packet is in the order of 243 bytes, and security relies on AES encryption.
- **Class A:** the device opens reception windows only after a transmission of its own. It is the lowest-consumption mode and is supported by all devices.

LoRaWAN: Architecture and Device Classes (2/2)

- **Class B:** adds periodically scheduled reception windows, synchronised by gateway beacons.
- **Class C:** keeps reception permanently open, except while transmitting, which minimises latency at the cost of much higher consumption.
- Typical applications include early earthquake detection and real-time traffic management.

LoRaWAN: Network Architecture



Topologia em estrela de estrelas: o dispositivo não se associa a um gateway específico

LoRaWAN: Device Classes

Janelas de receção por classe

Classe A duas janelas após cada envio	Classe B janelas periódicas sincronizadas	Classe C receção permanente
Consumo mínimo Latência descendente elevada	Compromisso intermédio	Consumo elevado Latência mínima

Sigfox: Overview (1/2)

- The first LPWAN technology to reach the market, with transmission based on BPSK modulation.
- The physical layer relies on ultra-narrowband wireless modulation, occupying very small portions of the spectrum and encoding data by changing the carrier phase.
- A proprietary system, operated under a subscription model, with a cloud platform and a Sigfox-defined API for customer servers to access the data.
- Very low data rate, in the order of 100 bit/s, with range reaching up to 50 km.

Sigfox: Overview (2/2)

- A limit of 140 messages per day per device, with a payload of about 12 bytes.
- It supports *roaming* but embeds no security mechanisms of its own at the technology level.

Sigfox in Portugal

- The Sigfox network was deployed nationally in 2014 by Narrownet, a Portuguese telecommunications infrastructure company.
- Network installation was completed in under one year.
- Coverage reached 92% of the population and 70% of the national territory.
- The network connected more than 142 thousand objects in Portugal.
- Partnerships were established with strategic customers, among them Janz and Securitas Direct.

Weightless: Overview

- A low-cost technology, designed to be embedded directly into machines.
- It operates in an unlicensed environment, where third-party interference is not predictable and must be avoided or compensated by the protocol itself.
- It uses narrowband spectrum, with range in the order of 13 km and a payload of about 200 bytes, protected by AES encryption.
- The Weightless Special Interest Group was created in 2012, associated with the release of spectrum in *white spaces*, ratified in the United States that same year.
- The first Weightless-N network was installed in London in 2014.

Weightless: Comparison of Variants

- **Weightless-N:** unidirectional communication, range above 5 km, battery life up to 10 years, very low terminal and network cost.
- **Weightless-P:** bidirectional communication, range above 2 km, battery life between 3 and 8 years, low terminal cost and medium network cost.
- **Weightless-W:** bidirectional communication, range above 5 km, battery life between 3 and 5 years, low to medium terminal cost and medium network cost.
- The choice between variants illustrates the recurring LPWAN trade-off: bidirectionality and range are paid for in battery life and infrastructure cost.

RPMA (Ingenu) (1/2)

- Random Phase Multiple Access is a low-power, long-range channel access method used exclusively for machine-to-machine communication.
- It operates in the 2.4 GHz band, widely available worldwide.
- It offers very broad coverage and high capacity, with good connection capacity per channel.
- It supports *handover*, that is, changing channel without dropping the connection.

RPMA (Ingenu) (2/2)

- It relies on Direct Sequence Spread Spectrum (DSSS), with convolutional channel coding and Gold codes for spreading, occupying 1 MHz of bandwidth.
- It uses TDD frames with both open-loop and closed-loop power control.

EnOcean: Battery-Free Communication (1/2)

- An ultra-low-consumption radio technology, based on miniaturised energy converters.
- The energy required for operation is harvested from the environment: motion, light or temperature, for instance the pressure applied to a switch or a photovoltaic cell.
- These sources are sufficient to power each module during transmission, which allows devices without any battery.

EnOcean: Battery-Free Communication (2/2)

- It operates at 868 MHz in Europe and 315 MHz in the United States.
- In 2014 the EnOcean Alliance gathered more than 300 members, among them Texas Instruments, Leviton, Osram, Sauter, Somfy, Wago and Yamaha.

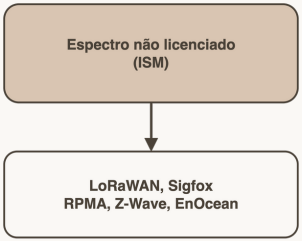
Z-Wave (1/2)

- A low-power radio protocol, aimed at home automation applications such as lighting and climate control.
- Low data rate, between 9 and 40 kbit/s, suited to occasional commands and readings.
- Devices may be battery-powered or mains-powered.
- It operates at 868 MHz in Europe and 908 MHz in the United States.
- Range is around 50 m, higher outdoors and lower inside buildings.

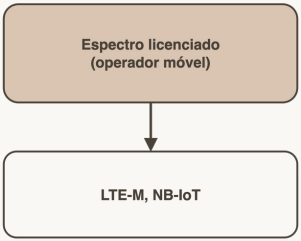
Z-Wave (2/2)

- It supports a mesh architecture, which extends coverage through forwarding between nodes.
- The medium access method is of the CSMA/CA type. The Z-Wave Alliance gathers more than 100 manufacturers.

Licensed and Unlicensed Spectrum



Sem custo de licença
Limites regulamentares de emissão
QoS não garantida



Gestão de interferência
Cobertura do operador
Custo por dispositivo superior

LTE-M: Overview (1/2)

- An evolution of LTE optimised for the Internet of Things, also designated eMTC.
- Introduced in 3GPP Release 12, in the fourth quarter of 2014, with optimisations in Release 13 and specifications completed in the first quarter of 2016. Commercially available since 2017.
- Reduced energy consumption and autonomous operation, with simplified deployment.
- It interoperates with existing LTE networks, which removes the need for dedicated infrastructure.
- Coverage up to about 11 km, with a maximum data rate of up to 1 Mbit/s.

LTE-M: Overview (2/2)

- It introduces a new terminal equipment category, Cat-0, of lower complexity and cost, with reception bandwidth reduced to 1.4 MHz and a power class of 20 dBm.
- The 15 dB gain in the link budget translates into better coverage and higher energy efficiency.

NB-IoT: Overview (1/2)

- It extensively reuses the LTE design, with FDMA in the downlink and SC-FDMA in the uplink.
- Lower cost than eMTC, as it operates in narrowband, with 180 kHz channels.
- Extended coverage: it supports a maximum coupling loss of 164 dB, against 144 dB for GPRS and 142.7 dB for conventional LTE.
- Reception sensitivity of -141 dBm and range in the order of 11 km, with a data rate close to 150 kbit/s.

NB-IoT: Overview (2/2)

- Battery life of up to 10 years with a 5 Wh battery, depending on traffic and coverage needs.
- It supports a massive number of devices: at least 50 000 per cell.

NB-IoT: Modes of Operation (1/2)

- The technology was designed with several deployment options over licensed GSM, WCDMA or LTE spectrum, so as to maximise the efficiency of spectrum use.
- **Stand-alone:** it occupies a dedicated carrier, typically obtained by reallocating GSM channels previously used by the GERAN system.
- **Guard band:** it occupies the unused resource blocks in the guard band of an LTE carrier.

NB-IoT: Modes of Operation (2/2)

- **In-band:** it occupies resource blocks within the LTE carrier itself, sharing it with conventional traffic.
- This flexibility allowed operators to introduce NB-IoT without investing in additional spectrum.

5G in Support of the Internet of Things (1/2)

- The IMT-2020 framework, defined by the International Telecommunication Union, establishes the requirements for fifth-generation networks.
- The concept of *network slicing* allows the same physical infrastructure to be logically divided into slices with distinct characteristics, each tuned to a type of service.
- One slice may be configured for massive machine communications, prioritising the number of connections and energy efficiency; another, for very low latency and high reliability communications.

5G in Support of the Internet of Things (2/2)

- This flexibility makes it possible to accommodate, on the same network, the conflicting requirements of different classes of IoT applications.
- Successive 3GPP releases realise this evolution incrementally, maintaining compatibility with earlier technologies.

Cost of Communication Modules (1/2)

- **LoRaWAN:** the Microchip module, with UART interface and proprietary stack, cost about 14.27 US dollars per unit and 10.90 dollars in batches of one thousand. The MultiTech module, XBee-compatible, was around 30 dollars per unit.
- **NB-IoT:** the Quectel BC95 module, compliant with 3GPP Release 13, with SIM/USIM interface and transmission at 100 bit/s, cost about 40 dollars. The Digi XBee Cellular NB-IoT module reached about 60 kbit/s downlink and 25 kbit/s uplink.

Cost of Communication Modules (2/2)

- The cost difference between families reflects terminal complexity and technology licensing.
- These figures are indicative and dated: they serve to compare orders of magnitude, not to budget a current solution.

Service Pricing Models (1/2)

- **NB-IoT:** in 2017, the Deutsche Telekom entry package started at 199 euros, including six months of activation for up to 25 SIM cards with 500 KB per SIM on the German network. The variant with access to the Cloud of Things platform started at 299 euros.
- In 2018 the first flat-rate data tariff for the Internet of Things appeared in Europe, in a joint offer from Deutsche Telekom and 1NCE, aimed at business customers.

Service Pricing Models (2/2)

- **Sigfox:** subscription in the order of one Singapore dollar per device per month, with a plan of up to 140 daily messages. Partners committing to volume could obtain figures close to one dollar per device per year.
- **Outcome Based Pricing:** an alternative model in which price derives neither from traffic volume nor from the number of devices, but from meeting objectives agreed between customer and supplier, which shares the investment risk between both.

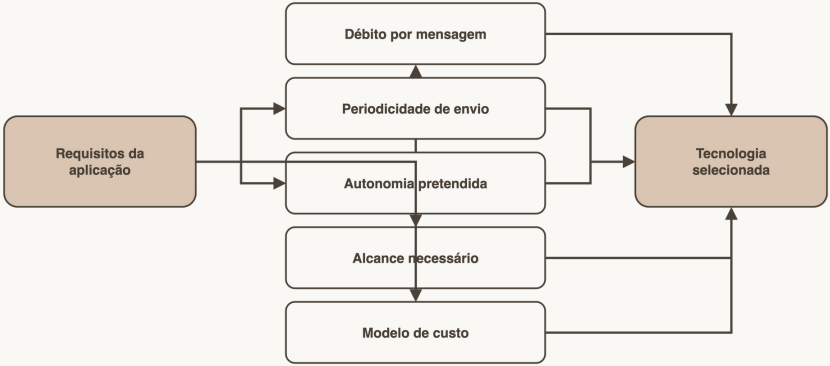
Evolution Prospects (1/2)

- According to Analysys Mason forecasts, NB-IoT will be the dominant IoT network technology in 2026.
- 3G and 4G technologies are expected to represent about 27% of market share on that horizon.
- 5G is expected to account for slightly more than 1% of total connections in 2026, averaged across all application groups.

Evolution Prospects (2/2)

- In the automotive segment and in embedded SIM cards, the 5G share will be higher, in the order of 4% of total connections.
- These forecasts confirm that the adoption of new generations in IoT is slower than in personal communications, owing to the long life cycles of installed equipment.

Criteria for Selecting an LPWAN Technology



Lesson Summary (1/2)

- LPWAN networks fill the space left between short-range technologies and conventional cellular networks, trading data rate for range and battery life.
- Non-3GPP solutions operate in unlicensed spectrum, with low operating cost but no quality-of-service guarantees: LoRaWAN offers bidirectionality and class flexibility, Sigfox favours simplicity and cost, and RPMA focuses on capacity and coverage.
- 3GPP solutions (LTE-M and NB-IoT) benefit from the installed cellular infrastructure and licensed spectrum, with better interference management at a higher cost.

Lesson Summary (2/2)

- EnOcean demonstrates that, in specific scenarios, it is possible to dispense with the battery entirely by harvesting energy from the environment.
- Selecting an LPWAN technology depends on the required data rate, message periodicity, intended battery life and acceptable cost model.

Sources

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Thank you.

Questions and comments